UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS



Doctoral Program in Deep Tech, Artificial Intelligence & Emerging Technologies

Research Methods and Scientific Integrity in AI and Advanced Technologies

REPRODUCIBILITY AUDIT REPORT OF A PUBLISHED LEGAL RETRIEVAL-AUGMENTED GENERATION STUDY
DELIVERABLE No. 6

Course Instructor	**Dr. Loveleen Gaur**
Student	**Eng. Jesús Vicente Inca Cárdenas**

2026 - II

Table of Contents

Executive Summary....3
1. Audited Paper and Scope....4
2. Experimental Setup....5
3. Reproducibility Scorecard....5
4. Detailed Audit and Evidence....6
5. Paper–Repository Consistency Check....8
6. Re-computation of the Reported Statistics....9
7. Missing Information and Reproducibility Risks....10
8. Overall Judgment....10
9. Three Priority Recommendations....11
10. Implications for My Doctoral Research....11
References....13
Appendix A. Evidence Map....14

REPRODUCIBILITY AUDIT REPORT OF A PUBLISHED LEGAL RETRIEVAL-AUGMENTED GENERATION STUDY
DELIVERABLE No. 6

Research topic: A Verifiable, Human-in-the-Loop Generative AI Model for Contractual Clause Recommendation in Regulated Fiduciary Institutions. Audited study: the closest published antecedent to the thesis, a legal RAG system evaluated on Peruvian labour law.

Audit scope. Evidence and ratings are based on the published article, a snapshot of its public code repository, and a re-computation of the reported statistics from the published tables. The LLM pipeline was not re-executed, and no participant- or question-level data were available.

Executive Summary

Jove-Ticona et al. (2026) propose a Metadata-Aware retrieval-augmented generation (RAG) architecture for legal question answering and evaluate it on a corpus of Peruvian labour law with 150 question–answer pairs, three open-weight LLMs (Llama-3.1-8B, GPT-OSS-20B and Gemma-3-27B) and four RAGAS metrics. They report double-digit gains over a naive RAG baseline in every metric and model (13.10% to 28.15%), a paired t-test with $t(11) = 15.49$ and Cohen's $d = 4.47$. I chose this paper because it is the closest published antecedent to my thesis: a legal RAG system, in Spanish, built and evaluated in Peru, and included as study A7 in my systematic review (Deliverable No. 4).

The study is transparent in several respects. It describes the hardware, the main hyperparameters and the construction of the evaluation set, reports a normality check, a paired test, a confidence interval and an effect size, and publishes a code repository that includes the legal corpus. Every statistic in the paper could be recomputed exactly from its Table II. The reproducibility picture is nevertheless incomplete. Each configuration was run once with stochastic decoding and no seed; the 150 evaluation pairs, the RAGAS evaluation scripts and the per-question results are not in the repository (the folders ragas/, scripts/ and results/ are excluded by .gitignore); several parameters in the code differ from those reported; and the inferential test treats 12 aggregated scores from three models and four metrics as independent observations.

Using the same transparent heuristic as previous course audits (Yes = 2, Partial = 1, No = 0), the paper receives **8/18 points (44.4%)**. The number is descriptive rather than psychometric. **Overall judgment: moderate methodological transparency and analytical reproducibility, low computational reproducibility, and limited evidence that the reported effect sizes would hold at the question level or outside the evaluation set.**

1. Audited Paper and Scope

1.1 Paper metadata

Table 1
Metadata of the Audited Paper

Field	Audited paper
Title	A New Metadata-Aware Retrieval-Augmented Generation (RAG) Architecture for Trustworthy Legal Question Answering
Authors	Alexandra V. Jove-Ticona; Luis J. Duarte-Coaquera; Israel N. Chaparro-Cruz; Silvana B. Cabana-Yupanqui; Americo Chaparro-Guerra (Universidad Nacional Jorge Basadre Grohmann, Tacna, Peru)
Journal	International Journal of Advanced Computer Science and Applications (IJACSA), 17(8), 741–751, 2026
DOI	https://doi.org/10.14569/IJACSA.2026.0170868
Study design	Controlled offline comparison of two RAG pipelines (naive vs. metadata-aware) × three LLMs, single evaluation set
Corpus	Political Constitution of Peru and three labour-law compilations (general, private regime, special regimes); more than 3,000 pages and 6,000 sub-chunks
Evaluation set	150 question–answer pairs generated by an LLM from random corpus chunks and validated by two legal experts
Primary outcomes	RAGAS Context Precision, Context Recall, Factual Correctness and Faithfulness
Primary analysis	Shapiro–Wilk test, one-tailed paired t-test, 95% CI of the mean difference, Cohen's d; linear regression of score on model size
Code and data	https://github.com/avjovet/Metadata-AwareRAG (snapshot audited: archive commit a0cda0fd, files dated 16 February 2026)

Note. Individual elaboration.

1.2 Audit scope and interpretation rule

The Session 6 scorecard asks about seeds and repeated runs, data splits, leakage, tuning separation, statistical tests, effect size and confidence intervals, external validation, compute documentation, and code and data availability. Unlike many legal-technology papers, this study is an ML evaluation in the full sense, so all nine items apply directly; only "splits" needs interpretation, because no model was trained and the 150 pairs act as a single test set.

Three sources of evidence were used. First, the article itself. Second, the public repository, which was inspected file by file to check whether the reported pipeline, parameters, evaluation set and evaluation code are available and consistent with the paper. Third, the reported statistics were recomputed from the published Table II with SciPy, to test analytical reproducibility. The pipeline was not re-run, because it requires local serving of three LLMs and, above all, the evaluation set that is not published.

2. Experimental Setup

The corpus combines the Political Constitution of Peru with three compilations of labour regulations published by the Ministry of Labour; the repository version of the private-regime compendium is dated 21 October 2024. The proposed pipeline has two components. Sub-chunking with legal metadata inheritance segments each instrument along its legal structure, extracts metadata (source, document type, title, topic, year and link) and splits it into sub-chunks of at most 250 tokens with a 30-token overlap, each inheriting the parent's metadata. The Adaptive Filter Creator corrects the query, routes it to a document category, extracts metadata filters, retrieves 20 candidates with progressive filter relaxation, and re-ranks them with jina-reranker-v2 to keep the top 5. The naive baseline uses fixed-length chunks, dense retrieval and the same k = 5.

Embeddings come from BAAI/bge-m3, vectors are stored in ChromaDB, and the LLMs are served locally with Ollama on a workstation with an Intel Core Ultra 9 285K, 128 GB RAM and an RTX 5090 (32 GB). Generation uses temperature 0.1 and top-p 0.9. The 150 evaluation pairs were produced by a generative model from randomly selected corpus chunks and retained only if the question was unambiguous and the answer appeared verbatim in one identifiable passage; two legal experts validated them, one from the research team and one external labour-law specialist. Each of the six pipeline × model configurations was evaluated once, and the mean RAGAS score over the 150 pairs is reported per metric (Table II of the paper). Statistical testing pools the 12 differences between pipelines (three models × four metrics).

3. Reproducibility Scorecard

Scoring rule: Yes = 2 points, Partial = 1 point, No = 0 points. The course material does not prescribe a numerical conversion, so this is an explicit heuristic that only summarises the item-level audit; the evidence and justification remain the primary result.

Table 2
Reproducibility Scorecard for Jove-Ticona et al. (2026)

Checklist item	Rating	Evidence from the paper and repository
Seeds reported, many runs?	NO	Each configuration was run once. Decoding is stochastic (temperature 0.1, top-p 0.9), no seed is reported or set in the code, and no run-to-run variance is given.
Splits described?	PARTIAL	No model is trained, so the 150 pairs are a single test set. Their construction and acceptance criteria are described, but the set is not released, the generating LLM is not named, and no development set separate from the test set is reported.
Leakage checked?	NO	No check is reported. The pairs were generated from corpus chunks (which segmentation is not stated), answers had to appear verbatim in a single passage, and one validator belongs to the research team; each can favour the proposed retrieval.
Tuning separated?	PARTIAL	Final hyperparameters are reported (250/30-token chunks, 20 candidates, top 5, τ = 0.65, T = 0.1, top-p = 0.9), but not how they were chosen or whether the 150 pairs were used while designing the pipeline.

Checklist item	Rating	Evidence from the paper and repository
Statistical test named?	YES	Shapiro–Wilk, a one-tailed paired t-test and a regression are named and fully reported. Validity concern: the 12 paired units are three models × four metrics, not independent samples (Section 4.5).
Effect size and CI?	YES	Cohen's d = 4.47 and a 95% CI of the mean difference [0.141, 0.187] are reported and reproduce exactly. No CIs are given per metric or per model, and d is computed on aggregated scores.
External validation?	NO	A single corpus, a single synthetic evaluation set, no real user queries, no held-out legal domain or later time period, no independent replication. The authors acknowledge this.
Compute documented?	PARTIAL	CPU, RAM, GPU, Python 3.10, frameworks and model names are documented; library versions, RAGAS version, Ollama model tags and quantisation, and run times are not.
Code and data available?	PARTIAL	Pipeline code and the legal corpus are public. Missing: the 150 pairs, the RAGAS evaluation scripts and per-question results (ragas/, scripts/ and results/ are git-ignored), pinned requirements, run instructions and a licence.

Note. Individual elaboration. Score: 2 Yes × 2 + 4 Partial × 1 + 3 No × 0 = 8/18 = 44.4%. Read as "partially reproducible under the course scorecard", not as a validated quality index.

4. Detailed Audit and Evidence

4.1 Seeds and repeated runs — NO

The paper reports a single evaluation per pipeline × model configuration. Generation uses temperature 0.1 with top-p 0.9 nucleus sampling, which is low but not deterministic, and neither the paper nor the code sets a decoding seed. Several stages of the proposed pipeline are themselves LLM calls (query correction, semantic routing, filter extraction), so sampling variation can change which documents are retrieved, not only the wording of the answer. Without repeated runs, the reader cannot tell how much of a 0.02 difference between two models is noise. Repeated runs with fixed seeds, or deterministic decoding, would be needed to separate model effects from run-to-run variation.

4.2 Data splits — PARTIAL

Because no model is trained, the relevant partition is between the data used to design the system and the data used to evaluate it. The paper describes the evaluation set well: 150 pairs, generated by an LLM from random chunks, kept only when clear, answerable from a single passage and verbatim, distributed across labour regimes and constitutional rights, and validated by two experts. Three gaps prevent independent reconstruction: the generating LLM and prompt are not named; the per-regime distribution is not tabulated; and no separate development set is reported, so it is unknown whether the same questions informed design decisions.

4.3 Leakage and contamination — NO

The paper does not report any check for leakage, and its design creates three plausible channels of advantage for the proposed system. First, the pairs were generated from "randomly selected chunks of the corpus"; if those were the metadata-rich sub-chunks of the proposed pipeline, the

gold evidence is aligned with its segmentation by construction. Second, the requirement that each answer appear verbatim in a single passage favours precise retrieval of that passage over reasoning, which is exactly what metadata filtering optimises; the authors acknowledge that the questions exclude multi-document reasoning. Third, one of the two validators is a member of the research team, and inter-annotator agreement is not reported. Finally, Context Precision is computed as cosine similarity between query and passage above $\tau = 0.65$, so it measures similarity to the question rather than relevance judged against a gold passage.

4.4 Tuning separation — PARTIAL

The final configuration is reported in useful detail: 250-token sub-chunks with 30-token overlap, 20 initial candidates, re-ranking to the top 5, $\tau = 0.65$, temperature 0.1 and top-p 0.9. The paper does not state how these values were chosen, whether alternatives were tried, or whether the 150 evaluation pairs were used during that process. The repository contains a chunking-comparison script, which suggests that configurations were compared during development. Without a statement that tuning used separate data, optimistic bias cannot be ruled out.

4.5 Statistical analysis — YES, with a validity concern

The tests are named and reported in full: a Shapiro–Wilk test on the paired differences ($W = 0.869$, $p = 0.064$), a one-tailed paired t-test ($t(11) = 15.49$, $p = 4.06 \times 10^{-9}$) and a linear regression of each metric on model size. All of these values were recomputed exactly from Table II (Section 6), so the analysis is analytically reproducible.

The concern is the unit of analysis. The 12 paired observations are three models × four metrics, each already averaged over the same 150 questions. They are not independent: the four metrics share the same retrieved contexts and answers, and the three models share the same retrieval pipeline. Treating them as 12 independent samples overstates the evidence. A test at the question level (150 paired questions per model and metric, with a paired bootstrap or a mixed-effects model, and correction for multiple comparisons; Dror et al., 2018) would be appropriate. The regression of score on model size rests on three points per line, so its p-values carry little information; it also uses 20B for GPT-OSS, while Table I lists 21B parameters.

4.6 Effect size and confidence intervals — YES, with limits

The paper reports the mean paired difference (0.164) with a 95% confidence interval [0.141, 0.187] and Cohen's $d = 4.47$, computed as the mean difference divided by the standard deviation of the 12 differences (the dz variant; Lakens, 2013). All three values reproduce. Two limits remain. The effect size inherits the unit-of-analysis problem, so $d = 4.47$ describes the consistency of 12 aggregated scores rather than the size of the improvement on individual questions. And no confidence intervals are given for the four metric-level or the three model-level differences, which are the comparisons a reader would act on.

4.7 External validation — NO

All results come from one corpus and one synthetic evaluation set, evaluated offline. There are no real user queries, no adversarial or ambiguous questions, no held-out legal domain, no later version of the norms and no independent replication. The authors list these limitations themselves and propose multi-jurisdictional and temporal-validity extensions as future work.

4.8 Compute documentation — PARTIAL

The hardware is documented precisely (Intel Core Ultra 9 285K, 128 GB RAM, RTX 5090 with 32 GB), as are Python 3.10, LangChain, HuggingFace, ChromaDB, Ollama and RAGAS, and the embedding, re-ranking and generation models. Missing are the versions of every library, the RAGAS version and the models it used as evaluators, the exact Ollama model tags and quantisation levels (which change model behaviour for the same nominal model), and run times. requirements.txt lists 15 packages without any version.

4.9 Code and data availability — PARTIAL

Publishing the code and the legal corpus is a real strength: the repository contains the pipeline modules, the source PDFs and the processed JSON chunks with their metadata. It does not contain what would be needed to reproduce the reported numbers: the 150 evaluation pairs, the RAGAS evaluation code, configuration files for the six runs, or per-question outputs. The .gitignore explicitly excludes ragas/, scripts/, results/ and the vector databases. The only experiment script, run_experiment.py, runs the naive pipeline on three hard-coded questions. There is no README with run instructions and no licence, which matters because the corpus includes a third-party compilation of labour law.

5. Paper–Repository Consistency Check

Comparing the paper with the code shows that the repository is not a frozen release of the evaluated system. Table 3 lists the differences found. None proves that the reported results are wrong, but each would make a re-run produce a different system from the one described.

Table 3
Differences Between the Paper and the Public Repository

Parameter	Paper	Repository
Generation temperature	T = 0.1 for all generative inference	Naive pipeline 0.1; metadata-aware pipeline and chatbot 0.0; global default 0.0
Top-p	0.9	Not set anywhere in the code
Sub-chunk size	250 tokens, 30-token overlap	config/settings.py: 512 and 30; JSON indexing default 1,200 and 100; comparison script 10,000 and 0
Retrieval depth	20 candidates, re-ranked to top 5	Metadata-aware default top_k = 15; settings DEFAULT_TOP_K = 10; re-ranker top_n = 5

Parameter	Paper	Repository
Embedding model	BAAI/bge-m3	config/settings.py: bge-m3; src/config/settings.py default: all-MiniLM-L6-v2 (two configuration modules)
Evaluation set	150 validated question–answer pairs	Not included; run_experiment.py has 3 hard-coded questions
Evaluation code	RAGAS with DeBERTa-large-MNLI, MiniLM and $\tau = 0.65$	Not included; ragas/, scripts/ and results/ are git-ignored
Dependencies	Python 3.10 environment	requirements.txt without versions; langchain_experimental used in a script but not listed

Note. Individual elaboration from the repository snapshot (archive commit a0cda0fd).

A related point concerns the metrics. The paper describes RAGAS as computing faithfulness and factual correctness with a DeBERTa-large-MNLI entailment model and context metrics with MiniLM cosine similarity. The RAGAS framework documents these metrics as LLM-based evaluations (Es et al., 2024), so the description suggests a customised configuration. Because the evaluation code is not released, the exact metric definitions cannot be verified.

6. Re-computation of the Reported Statistics

To test analytical reproducibility, the 12 paired differences were derived from Table II of the paper and the reported analyses were recomputed with SciPy. Every value matched to the reported precision (Table 4). This shows that the published numbers are internally consistent and that the inferential analysis can be audited from the tables alone, a positive finding that is uncommon in this literature.

Table 4

Reported Versus Recomputed Statistics

Statistic	Reported	Recomputed from Table II
Mean paired difference (n = 12)	0.164	0.164
SD of differences	0.0367	0.0367
Shapiro–Wilk	$W = 0.869$, $p = 0.064$	$W = 0.869$, $p = 0.064$
Paired t-test, one-tailed	$t(11) = 15.49$, $p = 4.06 \times 10^{-9}$	$t(11) = 15.49$, $p = 4.06 \times 10^{-9}$
95% CI of the mean difference	[0.141, 0.187]	[0.141, 0.187]
Cohen’s d (dz)	4.47	4.47
Improvement range	13.10% to 28.15%	13.10% to 28.15%
Faithfulness slope, proposed RAG	0.00480, $r = 0.991$, $p = 0.084$	0.00480, $r = 0.991$, $p = 0.084$ (with GPT-OSS = 20B)

Note. Individual elaboration. Computed with scipy.stats from the published Table II.

Analytical reproducibility is not computational reproducibility. The recomputation confirms the arithmetic from aggregated scores; it cannot confirm the scores themselves, which would require the evaluation set, the evaluation code and repeated runs.

7. Missing Information and Reproducibility Risks

Table 5
Missing Information and Why It Matters

Missing item or risk	Why it matters
The 150 evaluation pairs	Without them no independent team can recompute any metric; they are the single most important missing artefact.
Evaluation scripts and per-question scores	Needed to recompute RAGAS metrics, to test at the question level and to inspect failure cases.
Generating LLM and prompt for the evaluation set	The source model shapes the style and difficulty of the questions and may share biases with the evaluated LLMs.
Chunks used to generate the questions	If drawn from the proposed sub-chunks, the gold evidence is aligned with the proposed pipeline by construction.
Inter-annotator agreement of the two validators	The validity of the gold standard cannot be assessed; one validator is a co-author.
Seeds and repeated runs	With stochastic decoding and LLM-driven routing, single-run differences between models may be noise.
Pinned environment and model tags	Unversioned libraries, RAGAS and Ollama models will drift; quantisation changes model behaviour.
Frozen release matching the paper	Code parameters differ from the reported ones (Table 3); a tagged release would fix the evaluated version.
Corpus version and temporal validity	The compendia carry update dates (for example 21 October 2024), but the paper does not report a corpus snapshot date, and legal answers depend on which norms were in force.

8. Overall Judgment

Overall reproducibility judgment: PARTIAL — moderate methodological transparency and full analytical reproducibility, but low computational reproducibility and limited external validity.

The study is clear about its architecture, corpus, hardware, evaluation design and statistical procedure, discloses its limitations, and publishes code and corpus. Its statistics can be reproduced exactly from its own tables. Those are real strengths compared with much of the legal RAG literature screened in Deliverable No. 4.

What cannot be reproduced is the evaluation that produced the headline numbers. The evaluation set, evaluation code and per-question results are absent; the repository code differs from the reported configuration; each condition was run once with stochastic decoding; and the test pools non-independent aggregate scores. The design of the evaluation set may also favour the proposed retrieval. The paper therefore offers credible preliminary evidence that metadata-aware retrieval helps on verbatim, single-passage questions about Peruvian labour law, but not reproducible evidence of the size of that effect, nor of its behaviour on real queries, multi-passage legal reasoning or other legal domains.

9. Three Priority Recommendations

1. Release a frozen evaluation package

Publish the 150 question–answer pairs with their source passage identifiers, the generating model and prompt, the RAGAS version and evaluator configuration, the evaluation scripts, per-question scores for all six runs, pinned dependencies and Ollama model tags, under a tagged release with a permanent identifier and a licence. Reconcile the code defaults with the reported configuration, or provide the exact configuration file used for each run.

2. Fix the unit of analysis and quantify run-to-run variance

Run each configuration several times with fixed seeds, or with deterministic decoding, and report the variance. Test the pipeline difference at the question level, for each metric and model, with a paired bootstrap or a mixed-effects model with question as a random effect, and control the family-wise error across the twelve comparisons. Report confidence intervals for each metric-level difference rather than a single pooled effect size.

3. Build an independent gold standard and validate externally

Generate questions without access to the proposed segmentation, from source documents rather than chunks; include multi-passage and interpretive questions; have both validators external to the team and report inter-annotator agreement; and complement automatic metrics with human judgement of attribution and legal correctness. Then test on real user queries, on a later version of the norms and on a second legal domain.

10. Implications for My Doctoral Research

This audit is directly useful because the paper is my closest antecedent. It confirms that my contribution cannot rest on showing that a metadata-aware or verification-augmented RAG beats a naive RAG on automatic metrics, since this study already reports that for Peruvian labour law. What the audit shows is that the evaluation layer is where the field is weakest, and that is where my protocol from Deliverable No. 3 is designed to be stronger.

- **Gold standard.** My protocol requires lawyers external to the development team, a written guideline and Krippendorff's $\alpha \geq 0.67$. This audit shows why: a co-author validator without reported agreement weakens every downstream metric.
- **Independence of the evaluation set.** Questions and gold passages must be created without access to my own segmentation or metadata, otherwise the evaluation rewards alignment with my design rather than legal usefulness. I will add this as an explicit rule to the annotation guideline.
- **Attribution is not correctness.** Faithfulness computed automatically measures support by retrieved text, not legal correctness, which is the tension stated in Deliverable No. 1

(Rashkin et al., 2023; Wallat et al., 2025). My design keeps human attribution judgements and a separate correctness rubric.

- **Unit of analysis and repeated runs.** The protocol already specifies the drafting query as the unit, k repeated runs with fixed seeds and mixed-effects models; this audit gives a concrete example of how pooling aggregate scores inflates evidence.
- **A frozen, executable package.** The scaffold of Deliverable No. 5 (pinned environment, DVC, MLflow, manifest, automated stranger test) answers most of the gaps found here. Every reported parameter will be read from a single configuration file and logged with the run, so the paper and the code cannot drift apart.
- **Temporal validity.** Legal answers depend on which norms were in force. My knowledge base will record validity dates per provision and evaluate on a dated snapshot, as stated in the protocol.

References

Dror, R., Baumer, G., Shlomov, S., & Reichart, R. (2018). The hitchhiker's guide to testing statistical significance in natural language processing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 1383–1392). Association for Computational Linguistics. https://doi.org/10.18653/v1/P18-1128

Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2024). RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations* (pp. 150–158). Association for Computational Linguistics.

Jove-Ticona, A. V., Duarte-Coaquera, L. J., Chaparro-Cruz, I. N., Cabana-Yupanqui, S. B., & Chaparro-Guerra, A. (2026). A new metadata-aware retrieval-augmented generation (RAG) architecture for trustworthy legal question answering. *International Journal of Advanced Computer Science and Applications, 17*(8), 741–751. https://doi.org/10.14569/IJACSA.2026.0170868

Kapoor, S., & Narayanan, A. (2023). Leakage and the reproducibility crisis in machine-learning-based science. *Patterns, 4*(9), Article 100804. https://doi.org/10.1016/j.patter.2023.100804

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. *Frontiers in Psychology, 4,* Article 863. https://doi.org/10.3389/fpsyg.2013.00863

Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., Fox, E., & Larochelle, H. (2021). Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research, 22*(164), 1–20.

Rashkin, H., Nikolaev, V., Lamm, M., Aroyo, L., Collins, M., Das, D., Petrov, S., Tomar, G. S., Turc, I., & Reitter, D. (2023). Measuring attribution in natural language generation models. *Computational Linguistics, 49*(4), 777–840.

Wallat, J., Heuss, M., de Rijke, M., & Anand, A. (2025). Correctness is not faithfulness in retrieval augmented generation attributions. In *Proceedings of the 2025 International ACM SIGIR Conference on Innovative Concepts and Theories in Information Retrieval (ICTIR '25)*. ACM.

Appendix A. Evidence Map

Table A1

Location of the Evidence Used in the Audit

Audit issue	Source location	Evidence summary
Evaluation set	Paper, Section IV.E	150 pairs generated by an LLM from random chunks; verbatim single-passage answers; two validators, one internal
Decoding	Paper, Sections IV.E and IV.F	Temperature 0.1, top-p 0.9; single run; no seed
Hyperparameters	Paper, Section IV.F	250/30-token sub-chunks, 20 candidates, top 5, $\tau = 0.65$
Hardware and software	Paper, Section IV.A	Core Ultra 9 285K, 128 GB RAM, RTX 5090 32 GB; Python 3.10; LangChain, ChromaDB, Ollama, RAGAS
Metric implementation	Paper, Section IV.D	DeBERTa-large-MNLI, MiniLM, dependency parsing; cosine threshold for context metrics
Main results	Paper, Tables II and III	Gains of 13.10% to 28.15% in all metrics and models
Statistics	Paper, Tables IV–VI and Section V.D	Shapiro–Wilk, paired t-test on $n = 12$, 95% CI, Cohen's $d = 4.47$, regression on model size
Limitations	Paper, Section VI	Synthetic questions, no real queries, offline only, single legal domain
Code availability	Paper, Section IV.F; repository README	Public repository with code and corpus; no run instructions
Missing evaluation artefacts	Repository .gitignore	ragas/, scripts/, results/ and vector databases excluded
Parameter drift	Repository src/pipelines, config/ and src/config	Temperature, top-p, chunk size, top-k and embedding defaults differ from the paper
Corpus snapshot	Repository datajson_prepositional/	Private-regime compendium updated to 21 October 2024

Note. Individual elaboration.